

Strategic Financial Simulator: Implementation Report

This document details the technical process followed in Power BI Desktop to develop a dynamic financial simulator based on "What-if" parameters. The primary objective is to enable executive-level stakeholders to visualize projected revenue adjustments through a controllable price variable.

1. Data Model Configuration

To ensure precise and professional temporal analysis, the following foundations were established:

- **Data Cleaning:** The standard financial dataset was utilized, ensuring the integrity of revenue values and dates.
- **Calendar Table:** A dedicated date table was integrated to allow for chronological comparisons and logical sorting of months.

2. Parameter Development and DAX Measures

The core of the simulator lies in the ability to adjust variables in real-time. The following elements were implemented:

- **"What-if" Parameter:** A numeric parameter named Price Adjustment was created with a range from -0.5 to 0.5 (5% increments). This parameter automatically generates a slicer in the report.
- **DAX Projection Formula:** The Projected Revenue measure was developed to calculate the financial impact:

`Projected Revenue = [Total Revenue] * (1 + [Price Adjustment Value])`

3. Visualization and Scenario Analysis

The final report was structured to provide an intuitive reading of the data:

Visual Element	Functionality
Bar Chart	Compares <i>Total Revenue</i> (baseline) against <i>Projected Revenue</i> (projected) on a month-to-month basis.
Slicer (Slider)	Allows the user to slide to apply percentage price adjustments and observe immediate changes in the bars.
KPI Cards	Strategically placed to display key metrics such as total revenue and margins.

4. Resolution of Technical Challenges

Several critical configuration issues were resolved during construction:

- **Slider Synchronization:** Field assignments were corrected to ensure the visual object responded accurately to the parameter's numeric series.
- **Chronological Sorting:** The month column was configured so that the chart follows the order from January to December instead of alphabetical order.

The result is an interactive simulation tool ready for strategic decision support, demonstrating technical proficiency in DAX and interface design within Power BI.